Elements and Compounds

Watch. Predict. Explain.

Original revision guide with 100 practice questions and answer explanations.

Original teaching resource informed by KICD's revised 2024 Grade 8 curriculum. Five introductory missions for strand 1.0, sub-strand 1.1 Elements and Compounds. The curriculum allocates 18 lessons to this sub-strand; this is not a full replacement. Independent prototype, not KICD-endorsed. Teacher review required. The source is linked, never bundled.

KICD Integrated Science Grade 8, revised 2024. strand 1.0, sub-strand 1.1 Elements and Compounds. Source viewer pages 13-14 of 37. Newly written explanations, questions and visuals, not an official assessment or a copy of the source.

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Models and label reading only. No tasting, unknown chemicals, exposed wiring, mercury or lead handling. Do not mix or burn elements. Real practical work requires teacher approval.

Use the HTML animation first, try its prediction, then read the optional explanation. All 100 practice questions are below. Short answers are self-assessed; equivalent scientifically correct wording is acceptable. No personal health information should be entered.

Your learning route

1. Atoms, elements & molecules: Distinguish an atom, an element, a molecule and a compound.

2. Speak in symbols: Use the correct symbols and capitalisation for selected elements.

3. Word equations: Identify reactants and products in simple word equations.

4. Read a compound: Identify element types in selected two-element compounds.

5. Useful materials & labels: Connect elements and compounds with uses and safe label reading.

1. Atoms, elements & molecules

Distinguish an atom, an element, a molecule and a compound.

An element contains one type of atom. An atom is the smallest unit that retains an element's chemical identity. The coloured balls in our model stand for whole atoms; they are not photographs or drawn to scale.

A molecule has two or more atoms bonded together. An oxygen molecule contains oxygen atoms only, so it is still an element. A water molecule contains hydrogen and oxygen atoms, so water is a compound.

A compound contains different elements chemically combined in fixed proportions. Not all compounds consist of separate molecules: sodium chloride forms an extended ionic structure. Not all elements consist of isolated atoms.

A mixture contains substances physically combined rather than chemically joined in one fixed composition. A model's colour key represents atom identity; real atoms are not coloured classroom balls.

Visual mission

Same atoms, different groupings

A molecule can be an element or a compound.

Colours identify atom types; they are not literal atom colours or scale.

Think it through

Explain why oxygen molecules represent an element but water molecules represent a compound.

2. Speak in symbols

Use the correct symbols and capitalisation for selected elements.

An element symbol is a short international label. A one-letter symbol is capitalised. In a two-letter symbol, the first letter is a capital and the second is lowercase.

Useful examples are oxygen O, carbon C, hydrogen H, nitrogen N, iron Fe, aluminium Al and copper Cu. Silver is Ag, gold Au, chlorine Cl, sodium Na, mercury Hg and lead Pb.

Some symbols come from historical names, so they are not always the first letters of the English name. Cu is copper; C is carbon. Cl is chlorine, not the letters C and capital I.

An element symbol and a substance formula serve different purposes. O is the symbol for oxygen; O2 describes a molecule with two oxygen atoms. Learning the key prevents ambiguous labels.

Visual mission

Crack the symbol code

Letter case is part of the scientific label.

This is a symbol key, not a periodic-table simulation.

Think it through

Explain why CO and Co would not mean the same thing, even though only one capital letter changes.

3. Word equations

Identify reactants and products in simple word equations.

A word equation describes a chemical reaction using substance names. Reactants appear before the arrow; products appear after it. The arrow means 'forms' or 'produces', not 'is identical to'.

Examples include hydrogen plus oxygen forming water, carbon plus oxygen forming carbon dioxide, and copper plus oxygen forming copper oxide. Aluminium can react with oxygen to form aluminium oxide.

Sodium and chlorine can form sodium chloride. These reactants can be hazardous: this is a reading and modelling activity, not an instruction to mix or burn elements.

Atoms are rearranged in chemical reactions rather than created from nothing. Word equations name the substances; balancing symbolic equations is a separate task and is not required in this introduction.

Visual mission

Follow the atom count

Reactions rearrange atoms; they do not create or destroy them.

A diagram only, not a procedure for reacting gases. Word-equation reading does not require symbolic balancing.

Think it through

For carbon plus oxygen forming carbon dioxide, identify the reactants and product without performing the reaction.

4. Read a compound

Identify element types in selected two-element compounds.

Compound names and formulae help identify their elements. Water contains hydrogen and oxygen; carbon dioxide contains carbon and oxygen; sodium chloride contains sodium and chlorine.

Copper oxide contains copper and oxygen. Aluminium oxide contains aluminium and oxygen. These examples each contain two element types even when the total atom count is greater than two.

In the formula H2O, the small 2 means two hydrogen atoms per water molecule; the unnumbered O means one oxygen atom. CO2 represents one carbon and two oxygen atoms per molecule.

NaCl describes a one-to-one ratio in an extended ionic structure, not a separate sodium-chloride molecule. Chemical formulae are case-sensitive, and composition alone does not tell you whether a substance is safe.

Visual mission

Different types, fixed proportions

Count element types separately from the total number of atoms.

The salt diagram is a small part of an ionic structure, not isolated NaCl molecules.

Think it through

Explain the difference between 'two elements' and 'two atoms' using a water molecule.

5. Useful materials & labels

Connect elements and compounds with uses and safe label reading.

Copper is useful in electrical wiring because it conducts electricity well. Iron is widely used in structures, often as part of alloys such as steel. Aluminium is relatively light and is used in many frames and containers.

Gold and silver are used in jewellery and decorative objects. Their value, durability and appearance influence choices. A real object may be an alloy or coated material rather than a pure element.

Labels can identify ingredients and quantities. A food label may list a compound containing sodium rather than sodium metal. Read the name and context instead of assuming every mention of an element means the pure element is present.

Mercury and lead are hazardous. Learn their symbols from models and reliable information, not by opening equipment or handling unknown materials. Never taste laboratory samples, touch exposed wiring or take supplements because of a classroom exercise.

Visual mission

Choose a material for its job

A useful property explains a material's application.

Material examples only. Do not touch wiring or handle hazardous element samples.

Think it through

Choose a safe packaging label and explain one ingredient name without assuming it is a pure element.

Revision cards

Atom

A unit retaining an element's chemical identity.

Element

One type of atom.

Molecule

A discrete group of bonded atoms.

Compound

Different elements chemically combined in fixed proportions.

Mixture

Substances physically combined.

Symbol rule

Capital first letter; lowercase second letter where present.

Cu / C

Copper / carbon.

Reactants

Starting substances before the reaction arrow.

Products

Substances formed after the reaction arrow.

NaCl

A one-to-one ratio in an extended ionic structure, not a separate molecule.

100-question practice bank

40 multiple-choice and 60 short-answer questions. This is a practice bank, not a standardised exam.

Atoms, elements & molecules

1. An element contains how many types of atom?

A. Two in every case

B. None

C. An unlimited mixture

D. One

2. Which model represents a compound molecule?

A. Two oxygen atoms bonded

B. Unbonded oxygen atoms only

C. Hydrogen and oxygen atoms bonded together

D. One iron atom

3. Two bonded oxygen atoms form what?

A. A new element

B. A molecule of an element

C. A compound of two elements

D. A mixture of metals

4. Which is a compound?

A. Water

B. Pure copper

C. Oxygen

D. Gold

5. Which is a mixture?

A. Pure water

B. Pure gold

C. Pure sodium chloride

D. Air

6. What do coloured balls represent in this lesson?

A. Cells

B. Measured atom photographs

C. Whole atoms in a simplified model

D. Literal coloured planets

7. Which statement about compounds is accurate?

A. All are safe to taste

B. Not all consist of separate molecules

C. All contain one atom type

D. Every compound is a mixture

8. A water molecule has how many element types?

A. Two

B. One

C. Three

D. Zero

9. Define an atom.

10. Define an element.

11. Define a molecule.

12. Define a compound.

13. Explain why an oxygen molecule is not a compound.

14. Name the elements in water.

15. Why is air a mixture?

16. Can a molecule also represent an element? Give an example.

17. Why should sodium chloride not be pictured as isolated molecules without a caveat?

18. State one limitation of the ball model.

19. How does a mixture differ from a compound?

20. Does joining two atoms always create a new element?

Speak in symbols

21. What is the symbol for oxygen?

A. Ox

B. o

C. Og

D. O

22. What is the symbol for iron?

A. FE

B. F

C. Fe

D. Ir

23. Which symbol represents copper?

A. Co

B. Cu

C. C

D. CU

24. Which symbol represents sodium?

A. Na

B. So

C. NA

D. S

25. What is the symbol for gold?

A. Go

B. AU

C. Ag

D. Au

26. What is the symbol for silver?

A. AG

B. Au

C. Ag

D. Si

27. Which capitalisation is correct for chlorine?

A. cL

B. Cl

C. CL

D. cl

28. Which is the symbol for lead?

A. Pb

B. Le

C. PB

D. Ld

29. Write the symbol for carbon.

30. Write the symbol for hydrogen.

31. Write the symbol for nitrogen.

32. Write the symbol for aluminium.

33. Write the symbol for mercury.

34. Explain the difference between C and Cu.

35. Explain the difference between Ag and Au.

36. Correct the symbol FE for iron.

37. Correct the symbol na for sodium.

38. Why are standard symbols useful?

39. Does O2 replace O as the element's symbol?

40. What is the general capitalisation rule for two-letter symbols?

Word equations

41. Where are reactants written in a word equation?

A. After the arrow only

B. Only in a title

C. They are omitted

D. Before the arrow

42. What does the reaction arrow mean?

A. Is always safe to mix

B. Contains no atoms

C. Forms or produces

D. Is the same substance as

43. Hydrogen reacting with oxygen can form what?

A. Sodium chloride

B. Water

C. Pure copper

D. Gold

44. Carbon reacting with oxygen to make carbon dioxide has which product?

A. Carbon dioxide

B. Carbon only

C. Oxygen only

D. Copper oxide

45. Copper plus oxygen can form what compound?

A. Sodium chloride

B. Water

C. Aluminium oxide

D. Copper oxide

46. Sodium plus chlorine can form what?

A. Copper

B. Hydrogen

C. Sodium chloride

D. Carbon dioxide

47. What happens to atoms in an ordinary chemical reaction?

A. They stop being matter

B. They are rearranged

C. They are all destroyed

D. They turn into cells

48. Which activity is appropriate here?

A. Reading and modelling equations

B. Burning unknown powders

C. Mixing sodium with chlorine

D. Tasting products

49. Name the reactants when water forms from hydrogen and oxygen.

50. Name the product of carbon plus oxygen in the example.

51. Write a word equation for copper oxide formation.

52. Write a word equation for aluminium oxide formation.

53. Write a word equation for sodium chloride formation.

54. What is a product?

55. What is a reactant?

56. Does a word equation by itself give a safe laboratory method?

57. Why is a reaction arrow not simply an equals sign?

58. Which elements are present in carbon dioxide?

59. Must you balance symbolic equations to identify reactants in this lesson?

60. What does conservation mean in the atom model?

Read a compound

61. Which elements are in carbon dioxide?

A. Copper and oxygen

B. Carbon and sodium

C. Chlorine and gold

D. Carbon and oxygen

62. Which elements are in sodium chloride?

A. Sodium and copper

B. Nitrogen and oxygen

C. Sodium and chlorine

D. Silver and carbon

63. Which elements are in copper oxide?

A. Chlorine and oxygen

B. Copper and oxygen

C. Carbon and oxygen

D. Copper and hydrogen

64. A water molecule contains how many atoms in total?

A. Three

B. Two

C. One

D. Four

65. CO_2 has how many element types?

A. Three

B. One

C. Zero

D. Two

66. Which pair is in aluminium oxide?

A. Silver and oxygen

B. Iron and chlorine

C. Aluminium and oxygen

D. Aluminium and gold

67. NaCl is best described here as what?

A. A cell

B. A ratio in an ionic structure

C. A separate molecule of every salt

D. An oxygen element

68. Does knowing a formula guarantee safe handling?

A. No

B. Yes, all compounds are safe

C. Only if it has two letters

D. Only if it is white

69. Identify both element symbols in H_2O.

70. Name the elements in aluminium oxide.

71. Explain the 2 in H_2O.

72. How many oxygen atoms are shown in one CO_2 molecule?

73. How many carbon atoms are shown in one CO_2 molecule?

74. Why does water have two element types but three atoms per molecule?

75. What element does oxide indicate in these compound names?

76. Can a compound have properties different from its elements?

77. Why should a formula's capital letters be preserved?

78. What does the Na-to-Cl ratio in NaCl indicate?

79. Why is a name not an invitation to taste a sample?

80. What is meant by a two-element compound?

Useful materials & labels

81. Why is copper commonly used in wiring?

A. It is an electrical insulator

B. It is always the cheapest substance

C. It contains no atoms

D. It conducts electricity well

82. Which element is often used in jewellery?

A. Sodium metal

B. Oxygen gas

C. Gold

D. Chlorine gas

83. Which element is a major part of ordinary steel?

A. Gold

B. Iron

C. Mercury

D. Chlorine

84. Which is a safe way to study mercury?

A. Use models and reliable information

B. Break open equipment

C. Touch a spill

D. Taste it

85. A label lists sodium chloride. Does that mean it contains sodium metal?

A. Always

B. Only if the packet is white

C. Only if it is expensive

D. No

86. What should you do with an unknown laboratory sample?

A. Smell it directly

B. Mix it with everything

C. Follow the teacher's safety instructions

D. Taste it

87. Why might aluminium be chosen for a frame?

A. It cannot form compounds

B. It is relatively light

C. It has no mass

D. It is a gas at room temperature

88. What can a packaging label help identify?

A. Ingredients and stated quantities

B. Every possible safety fact

C. The exact position of every atom

D. A guarantee of purity

89. Give one use of copper.

90. Give one use of iron-containing materials.

91. Give one use of gold or silver.

92. Why may a metal object not be a pure element?

93. What should you read before using a packaged product?

94. Explain why sodium in a food label is not an instruction to handle sodium metal.

95. Name two hazardous elements mentioned in this topic.

96. Why must exposed wiring not be used as a classroom sample?

97. What property helps explain copper's use in electrical connections?

98. Does a valuable material always make the best tool?

99. Why should a lesson not tell learners to take a mineral supplement?

100. How can learners share label findings safely?

Answer key & explanations

1. D. One

An element is defined by one atom type.

2. C. Hydrogen and oxygen atoms bonded together

A compound contains different elements chemically combined.

3. B. A molecule of an element

Bonding two atoms of the same element does not make a compound.

4. A. Water

Water contains hydrogen and oxygen chemically combined.

5. D. Air

Air contains several substances physically mixed.

6. C. Whole atoms in a simplified model

Model colours are an identification key.

7. B. Not all consist of separate molecules

Ionic compounds can form extended structures.

8. A. Two

Hydrogen and oxygen are the two types.

9. The smallest unit retaining an element's chemical identity.

An atom is not the same as a cell.

10. A substance containing one type of atom.

Its atoms may be bonded to others of the same type.

11. Two or more atoms bonded together as a discrete group.

A molecule can contain one or more element types.

12. Different elements chemically combined in fixed proportions.

Physical mixing alone does not make a compound.

13. It contains only oxygen atoms.

A compound requires different elements.

14. Hydrogen and oxygen.

Water is not a new element.

15. Its different gases are physically combined rather than all bonded into one compound.

Its composition can vary.

16. Yes, an oxygen molecule.

All its atoms belong to the same element.

17. It forms an extended ionic structure.

A formula unit gives a ratio, not a separate molecule.

18. It is not to scale and the colours are only a key.

Models select features rather than showing everything.

19. A mixture is physically combined; a compound has chemically combined elements in fixed proportions.

The distinction is not just appearance.

20. No.

Element identity depends on atom type, not simply bonding.

21. D. O

The selected element's symbol is a capital O.

22. C. Fe

The second letter must be lowercase.

23. B. Cu

C is carbon; Cu is copper.

24. A. Na

Some symbols reflect historical names.

25. D. Au

Ag is silver, not gold.

26. C. Ag

Silver's standard symbol is Ag.

27. B. Cl

Use a capital first letter and lowercase second letter.

28. A. Pb

Lead uses the historical symbol Pb.

29. C.

It is a one-letter capital symbol.

30. H.

Do not replace it with an invented abbreviation.

31. N.

The symbol is case-sensitive.

32. Al.

The second character is lowercase l.

33. Hg.

Mercury is hazardous; a symbol exercise does not require handling it.

34. C represents carbon; Cu represents copper.

Capitalisation and the second letter matter.

35. Ag is silver and Au is gold.

Similar-looking symbols can identify different elements.

36. Fe.

The second letter is lowercase.

37. Na.

The first letter is uppercase.

38. They provide concise, internationally understood element labels.

Invented local abbreviations could cause confusion.

39. No. O is the symbol; O2 is a molecular formula.

The number describes atoms in the molecule.

40. Capital first letter, lowercase second letter.

Changing case can change meaning.

41. D. Before the arrow

Reactants are the starting substances.

42. C. Forms or produces

It indicates the direction of the described reaction.

43. B. Water

Water contains hydrogen and oxygen.

44. A. Carbon dioxide

The product is named after the arrow.

45. D. Copper oxide

The compound contains copper and oxygen.

46. C. Sodium chloride

The product combines sodium and chlorine.

47. B. They are rearranged

The model conserves atom types and numbers.

48. A. Reading and modelling equations

These examples are not practical instructions.

49. Hydrogen and oxygen.

These are the starting substances.

50. Carbon dioxide.

The named product is after the arrow.

51. Copper + oxygen -> copper oxide.

This describes substances, not an experimental method.

52. Aluminium + oxygen -> aluminium oxide.

The product contains aluminium and oxygen.

53. Sodium + chlorine -> sodium chloride.

Do not attempt this hazardous reaction.

54. A substance formed in a reaction.

It is written after the arrow in the word equation.

55. A starting substance in a reaction.

It is written before the arrow.

56. No.

Safety, equipment and supervision require separate assessment.

57. It describes a transformation into products.

Reactants and products have different chemical identities.

58. Carbon and oxygen.

Their atoms are rearranged into the product.

59. No.

Naming reactants and products is the introductory goal.

60. The reaction rearranges atoms without losing or creating them.

A useful model must keep its atom accounting consistent.

61. D. Carbon and oxygen

The name identifies carbon and oxygen.

62. C. Sodium and chlorine

Chloride in this name relates to chlorine.

63. B. Copper and oxygen

Oxide indicates oxygen combined with another element.

64. A. Three

Two hydrogen atoms and one oxygen atom make three.

65. D. Two

C and O identify two types, despite three atoms per molecule.

66. C. Aluminium and oxygen

The two-element compound's name supplies the pair.

67. B. A ratio in an ionic structure

Its formula is not a claim of isolated NaCl molecules.

68. A. No

Safety depends on hazards, exposure and context.

69. H and O, representing hydrogen and oxygen.

Element symbols identify composition, not safe handling.

70. Aluminium and oxygen.

It contains two element types.

71. There are two hydrogen atoms per water molecule.

The number applies to the preceding element symbol.

72. Two.

The subscript follows O.

73. One.

An omitted number means one in this molecular formula.

74. Its molecule has two hydrogen atoms and one oxygen atom.

Atom count and element-type count answer different questions.

75. Oxygen.

The other part of the name identifies the other element.

76. Yes.

Combining chemically can produce a substance with very different properties.

77. They distinguish element symbols and prevent misreading.

Chemical notation is case-sensitive.

78. A one-to-one ratio.

It does not mean a separate NaCl molecule.

79. A known name does not remove its hazards.

Laboratory samples must never be tasted.

80. A compound containing two different element types.

It can contain more than two atoms in a molecular unit.

81. D. It conducts electricity well

Material properties help explain uses.

82. C. Gold

Gold's appearance and durability are useful properties.

83. B. Iron

Steel is an alloy, not usually pure iron.

84. A. Use models and reliable information

Mercury can be hazardous.

85. D. No

A compound is not the same as the separate pure element.

86. C. Follow the teacher's safety instructions

Unknown samples may have hidden hazards.

87. B. It is relatively light

Different applications require different properties.

88. A. Ingredients and stated quantities

Read labels in context and follow safety information.

89. Electrical wiring.

Good electrical conductivity supports that use.

90. For example, building structures or tools made of steel.

The object may be an alloy rather than pure iron.

91. Jewellery or decorative objects.

Appearance and durability affect material choice.

92. It may be an alloy, coating or combination of materials.

Appearance alone cannot establish composition.

93. Its instructions, ingredients and safety information.

A single chemical name is not the whole context.

94. The label may refer to sodium in compounds, not the pure reactive metal.

Chemical form changes properties and hazards.

95. Mercury and lead.

Study them without handling samples.

96. It can cause electric shock or other injury.

A material-use discussion is not a practical wiring task.

97. Good electrical conductivity.

The chosen property should relate to the intended use.

98. No.

Function, safety, strength and cost also matter.

99. Supplement decisions require appropriate health guidance, not a symbol exercise.

Learning labels is not medical advice.

100. Share an explanation without personal details or unsafe handling.

Photograph only appropriate public packaging if permitted.

Explore responsibly

The HTML includes YouTube topic-search links, not teacher-vetted individual videos. External content may include advertising, inaccurate claims or age-inappropriate recommendations. A teacher should preview results. The original built-in animations need no internet.

Share the complete site folder, or host it with index.html at the root. Progress is stored locally and separately for each topic; there are no accounts or shared class records.

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